

## SYNOPSIS

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**Name:**

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**Project Title:** Red ELISAR: A Privacy-Preserving Autonomous Red Team Agent Using RAG and Edge AI

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### Abstract

Red ELISAR (Edge LLM Integrated Security Agent for Red Teaming) is a locally deployable Retrieval-Augmented Generation (RAG)-based offensive security system developed to automate realistic attack-chain generation for penetration testing. The project addresses major limitations of standalone large language models in cybersecurity, especially hallucination and weak domain grounding, by constraining all generated attack steps to retrieved MITRE ATT&CK knowledge. In the implemented pipeline, more than 770 ATT&CK techniques are embedded using a sentence-transformer model and indexed in a FAISS HNSW vector store for low-latency semantic retrieval. For a given target scenario, the system retrieves top-ranked relevant techniques and uses a local Mistral model to generate structured multi-step attack chains aligned to those techniques. A faithfulness validation module verifies cited technique IDs against the retrieved context to detect unsupported references. To enable practical evaluation, the project includes a purposely vulnerable Flask web application and an integrated vulnerability red agent that scans discovered weaknesses and feeds contextual findings into attack-chain generation. A command-line red agent interface was also developed for reproducible testing and analyst-driven execution. Comparative benchmarking against non-RAG baselines shows strong improvements in factual grounding, coherence, and reliability, with observed hallucination near zero and high valid-technique alignment, while maintaining end-to-end runtime suitable for near real-time local operation (around 7–8 seconds). The work demonstrates that a retrieval-grounded local red-teaming architecture can produce safer, explainable, and operationally useful offensive simulations without cloud dependency.

### Specific Contribution

- Implemented local Mistral-based structured attack-chain generation.
- Developed faithfulness validator for ATT&CK technique-ID verification.
- Built benchmarking framework against baseline non-RAG models.

- Measured and reported performance metrics (grounding, coherence, latency).

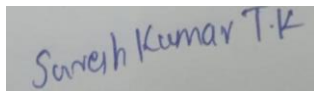
### **Specific Learning**

- LLM behavior control through constrained generation.
- Automated validation methods for hallucination reduction.
- Security evaluation metrics and experimental reporting.

### **Technical Limitations & Ethical Challenges Faced**

- Generated chains still depend on quality of retrieved context.
- Benchmark fairness and prompt sensitivity across model baselines.
- Ethical responsibility in presenting offensive outputs for defense-only use.

**Keywords:** *LLM Generation, Faithfulness Validation, Benchmarking, Mistral, Hallucination Reduction*



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